

BRUGGEN, Germany

WMO: 104010

Lat: 51.200N

Lon: 6.133E

Elev: 76

StdP: 100.42

Time Zone: 1.00 (EUC)

Period: 92-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.4	-11.4	1.4	-5.4	-8.9	1.8	-3.9	12.1	8.5	10.9	8.1	2.9	70	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.2	19.5	28.1	18.7	26.0	18.1	20.9	27.3	19.9	26.0	18.9	24.4	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	23.4	17.7	12.8	22.2	16.8	12.1	21.2	60.6	27.2	57.0	26.5	53.8	24.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	-10.9	33.5	3.2	2.1	-13.2	35.0	-15.1	36.2	-16.9	37.4	-19.2	38.9		
(5)			-11.6	23.1	3.0	1.5	-13.7	24.2	-15.5	25.1	-17.2	26.0	-19.4	27.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	3.1	3.8	6.4	9.6	14.2	16.2	18.5	18.5	14.7	10.2	5.9	3.5		
	DBStd	6.72	4.43	3.99	3.23	3.80	3.85	3.43	3.34	3.41	2.70	3.36	3.92	4.17		
	HDD10.0	941	216	174	119	53	7	0	0	0	1	39	128	205		
	HDD18.3	3011	474	406	371	264	136	84	39	40	114	251	372	461		
	CDD10.0	1095	1	1	7	39	138	186	263	264	141	47	6	3		
	CDD18.3	124	0	0	0	1	9	19	44	46	4	0	0	0		
	CDH23.3	1162	0	0	0	10	101	199	416	409	26	1	0	0		
	CDH26.7	342	0	0	0	0	17	52	131	139	4	0	0	0		
(14) Wind	WSAvg	3.3	3.9	3.7	3.7	3.2	3.1	3.0	2.9	2.7	2.9	3.1	3.3	3.8		
(15) Precipitation	PrecAvg	761	63	59	50	44	58	72	75	79	59	62	67	73		
	PrecMax	974	136	137	93	103	112	189	163	162	159	154	108	157		
	PrecMin	597	6	10	3	1	17	14	32	18	12	34	6	26		
	PrecStd	95	30	30	25	23	26	37	35	43	37	24	25	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.1	17.8	25.1	28.9	31.1	32.8	32.9	26.5	21.4	15.8	13.5		
		MCWB	10.4	9.7	11.1	13.1	17.5	20.9	21.1	21.1	18.2	16.6	12.0	11.0		
	2%	DB	11.4	12.1	15.5	21.4	25.9	27.9	30.1	30.4	23.1	18.6	13.4	12.3		
		MCWB	9.3	8.7	10.0	13.3	16.6	18.2	19.9	19.5	17.2	15.1	11.5	10.5		
	5%	DB	10.1	10.7	13.2	18.3	24.0	25.6	27.7	27.9	21.0	16.7	12.4	11.0		
		MCWB	8.4	8.4	9.3	12.0	15.4	17.5	19.1	19.0	15.8	13.8	10.7	9.4		
	10%	DB	8.9	9.4	11.5	15.7	21.7	23.1	25.5	25.2	19.2	15.3	11.0	9.7		
		MCWB	7.3	7.6	8.8	10.7	14.7	16.3	18.5	18.3	15.0	13.0	9.5	8.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.5	12.2	15.8	19.2	21.1	22.6	22.3	19.5	17.3	13.1	11.9		
		MCDB	12.4	12.8	15.7	21.5	26.4	28.3	29.7	30.4	24.2	20.6	14.0	12.7		
	2%	WB	9.6	9.5	10.9	13.7	17.5	19.6	21.1	20.8	17.8	15.7	11.8	10.7		
		MCDB	11.0	11.3	13.6	18.9	24.1	26.3	27.4	27.1	21.7	17.8	13.1	12.1		
	5%	WB	8.6	8.8	10.0	12.6	16.3	18.3	20.0	19.7	16.8	14.5	10.9	9.6		
		MCDB	10.0	10.4	12.3	17.0	21.8	23.8	26.3	25.6	20.0	16.2	12.2	10.8		
	10%	WB	7.4	7.7	9.2	11.5	15.2	16.9	18.9	18.7	15.8	13.3	9.7	8.3		
		MCDB	8.7	9.3	11.3	15.2	20.3	21.4	24.3	23.9	18.6	14.9	10.7	9.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.8	7.2	8.6	9.7	9.6	9.6	9.9	7.8	7.0	5.3	4.6		
		MCDBR	6.5	7.0	10.4	12.1	13.3	13.3	13.3	13.7	10.4	8.8	6.2	5.6		
	5% WB	MCWBR	5.1	4.8	6.3	5.8	6.0	6.0	5.5	5.4	5.3	5.6	4.8	4.8		
		MCWBR	6.3	6.2	8.1	10.4	12.0	11.7	12.4	11.7	9.2	8.0	5.7	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.348	0.378	0.413	0.418	0.418	0.425	0.416	0.388	0.362	0.338	0.321		
		taud	2.403	2.365	2.285	2.213	2.238	2.255	2.236	2.285	2.353	2.417	2.404	2.372		
	Ebn at Noon		691	772	812	824	837	840	825	813	794	746	665	628		
		Edh at Noon	62	83	107	128	131	130	131	118	99	78	61	55		
(44) All-Sky Solar Radiation	RadAvg		0.74	1.42	2.56	4.10	4.92	5.33	5.05	4.31	3.17	1.87	0.88	0.57		
	RadStd		0.11	0.27	0.38	0.46	0.44	0.47	0.62	0.41	0.35	0.17	0.11	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page